



Beyond Searching a Village: Learning to Recommend Diverse and Successful Collaborative Teams

<https://fani-lab.github.io/OpeNTF/tutorial/recsys26/>



fani-lab.github.io



recsys.acm.org/recsys26/



TIME AND LOCATION

ABSTRACT

TARGET AUDIENCE

OUTLINE

PIONEERING TECHNIQUES

NEURAL TEAM RECOMMENDATION

TIME AND LOCATION

Half-day, Afternoon, 1:30 PM - 5:30 PM (GMT/UTC -6), Monday, September 28, 2026

Elk/Deer Lake (4th floor), Marriott City Center, Downtown, Minneapolis, Minnesota, USA

Prev Tutorials at UMAP24, SIGIR-AP24, and WSDM25, and CIKM25 < [Full Outline] [Slides] [Recording]

Noah 4 Saanich



Forward-thinking for a better future

Noah Fruttarol

Candidate for Saanich Councillor

Who am I?

I am a University of Victoria Computer Science student, environmentalist, urbanist, and long-time Saanich resident. I'm dedicated to bringing a fresh, practical perspective to the Saanich Council. My candidacy is rooted in the lived realities of our district, having navigated our tough housing market, transit system, and the daily financial pressures facing all of us.

If you have any questions or want to discuss anything, call me anytime!!

(672)-381-3912

noahfruttarol.ca Follow Me!!

Noah4Saanich



• **Affordability:**

- Support non-market housing incentives, increasing supply in urban areas, around UVic, and local transit hubs.

• **Commutability:**

- Improving access to all types of transit.

• **Livability:**

- Support forward-thinking planning to create walkable neighbourhoods while improving access to green and community spaces (such as parks and recreation centers).

• **Responsibility:**

- I promise to be approachable and not just stand by my decisions, but justify them.

Authorized by Blake Baldwin, financial agent,
baldwin.bc.fm@gmail.com

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Related Work](#)

1

Team Projects List ▾

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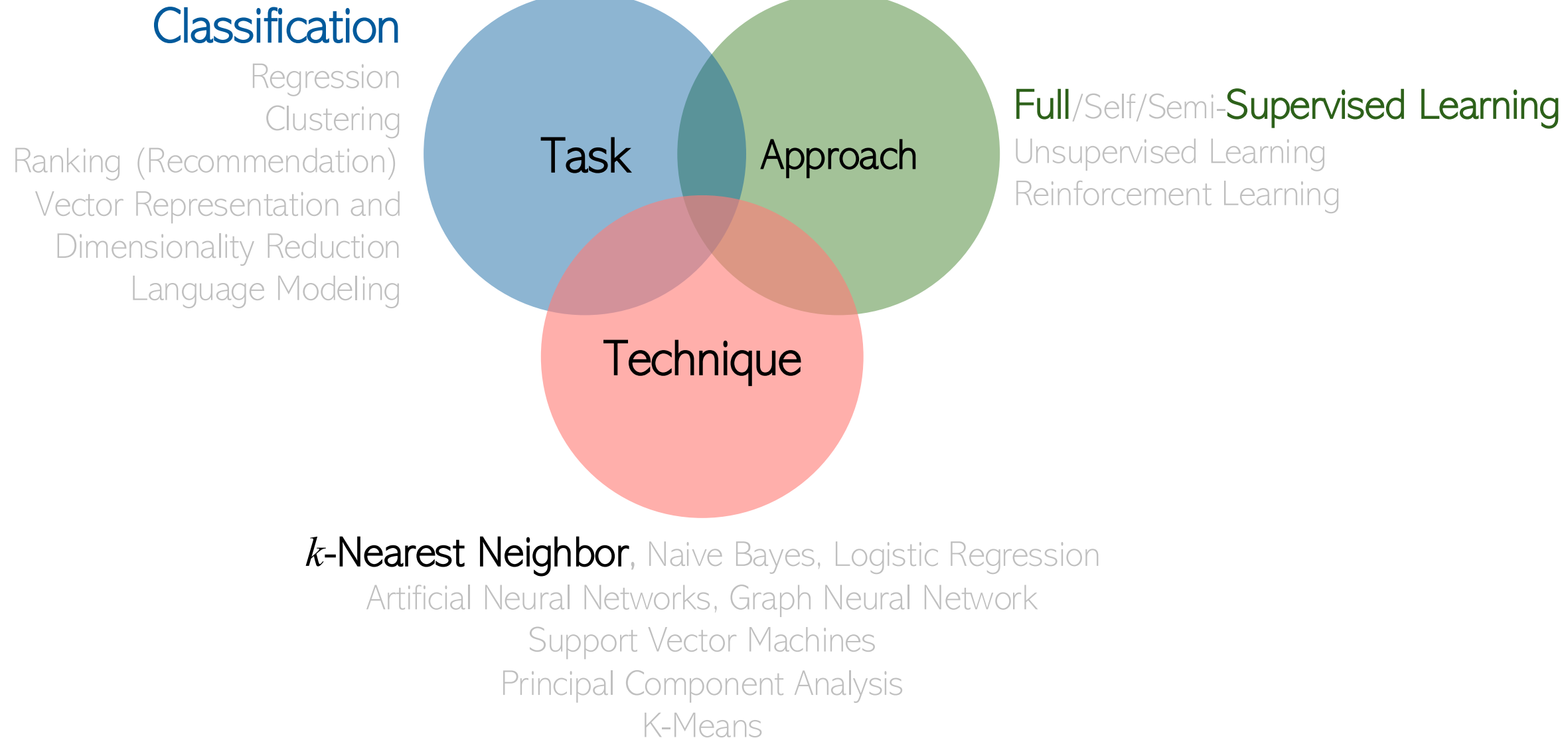
Add dates and restrictions...



A: Ali, R: Reia

Team ID	Student	Project Title	K
1A	Mason McNeill	Identifying Manipulative Twitter Bots Through Metadata Analysis	S
	Ken Wright		F
	Arunkumar Krishnamoorthy		T
	Carter Dunstan		M
10A	Douglas MacGregor	Image-Based Classification of British Columbia Seaweed	C
	Arda Berklin		S
	Jarren Morris		M
			In
1R	Alicia Kim	Understanding URL Characteristics Behind AI-Generated Phishing Evasion	K
	Jay Cheng		B
	Anson Koh		S
	Owen Pritchard		P
2R	Abhishek Majumdar	PRISM: Evaluating Privacy-Constrained Customer Churn Classification	M
	Dev Parsana		A
	Ravi Vashi		
	Bharath Irukulapati		
3R	Hung Vuong	MajorMatch: From Hobbies to Major	
	Brendan Wong		
	Robert Legassicke		
	Yashwanthkrishna Nagharaj		

4 Outline: W02B: Topics Start



5 Outline: W03A: Topics Start

- I) Task: Classification
- II) Supervision: Fully Supervised
- III) Technique: k -Nearest Neighbor
Let's zoom in ... Hyperparameters
- III) Evaluation

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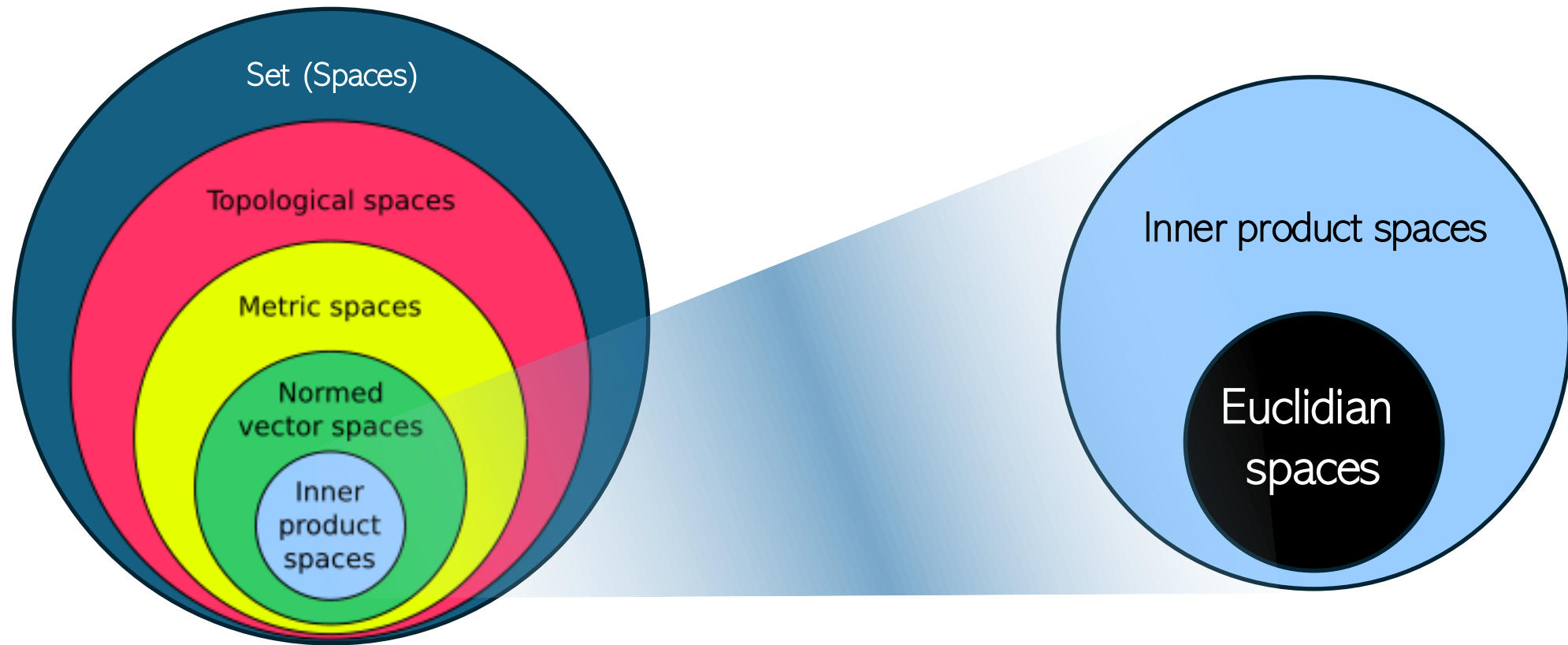
 k -Nearest Neighbor: Distance Function

$$f(\hat{o} : \mathbf{O}^*) = \underset{c_i \in \mathbf{C}}{\operatorname{ARGMAX}} \left\{ \sum_{\substack{(o,c) \in \operatorname{ARGSORT}\{d(\hat{o}, o)\} \\ (o, c) \in \mathbf{O}^*}}^k 1 [c_i = c] \right\}$$


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k -Nearest Neighbor: Space Hierarchy

We need a space that covers observations for defining distance function.



“nice geometric space”

yoo-KLID-ee-uhn

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k -Nearest Neighbor: Domain Challenges

How to map a real-world observation to a point in Euclidian space?

People

Animals (Birds, ...)

Objects (Houses, Cars, ...)

Image (Biometric Iris)

Audio (Voice, Music)

Video (Movies)

Text (Words, Sentence, Docs)

Timeseries

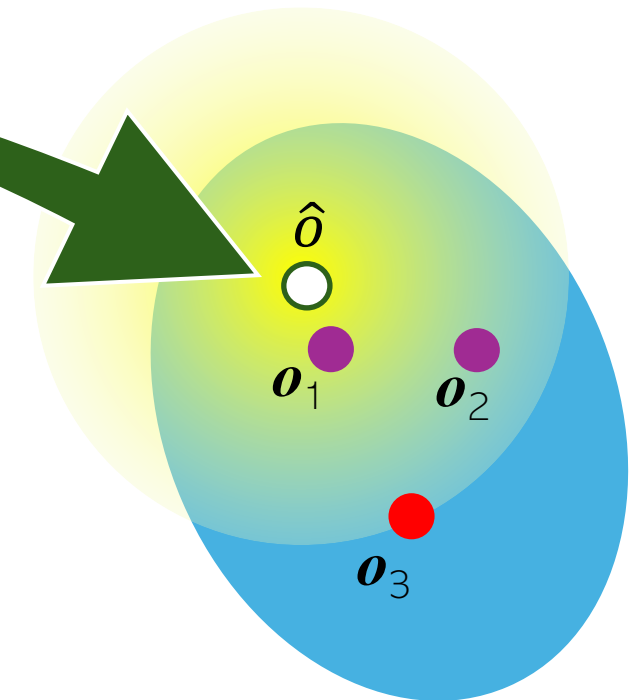
Genes

Spatial (GIS)

Light Bulbs

...

Encoding
Embedding
 $e(o)$

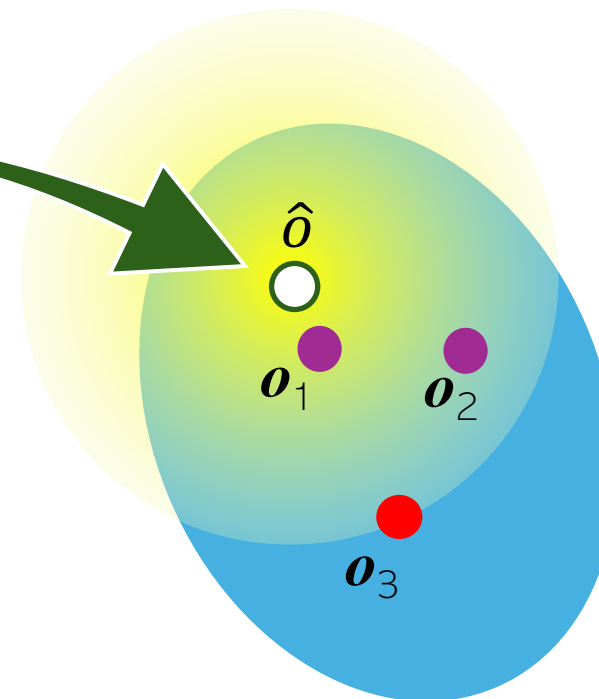


Euclidian Space
(Embedding Space)

9 k -Nearest Neighbor: Text

Words
Sentences
Docs
Social Posts
Books
Webpages
...

$e(o)$



Euclidian Space
(Embedding Space)

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k -Nearest Neighbor: Text: Word Space $\rightarrow$ Euclidian Space

Real-World Problem: Named Entity Annotator

Task: Classification $\mathbf{C} = \mathbf{Entities} = \{\text{Person, Organization, Event, Food, Animal}\}$

Approach: Fully Supervised $\mathbf{O}^* = \{(w, c)\}$, $\mathbf{U}^* = \{(\hat{w}, f(\hat{w}))\}$

Technique: $f: k\text{-NN} \rightarrow d(\hat{w}, w) \approx d(e(\hat{w}), e(w)) \rightarrow e: \mathbf{U} = \mathbf{Vocabs} \rightarrow \mathbb{R}^{n=1}$

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k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=1}$

Indexing $\rightarrow$ Enumerate words in a language's dictionary (vocabulary set)



$$e(\hat{w}: \text{'aardvark'})=0$$



$$e(w: \text{'apple'})=1050$$



$$e(w: \text{'zyzzyva'})=|V|$$

Does $e(.)$ preserve “real-world” proximity to “spatial” proximity? **No!**

$$\begin{aligned} |e(\text{'aardvark'}) - e(\text{'apple'})| &< |e(\text{'aardvark'}) - e(\text{'zyzzyva'})| \\ |0 - 1050| &< |0 - 520,000| \end{aligned}$$

'aardvark' is closer to 'apple' vs. 'zyzzyva'!

'aardvark' is tech company!

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k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

How to **define** meaning of a word? **We cannot!** E.g., “*game*”

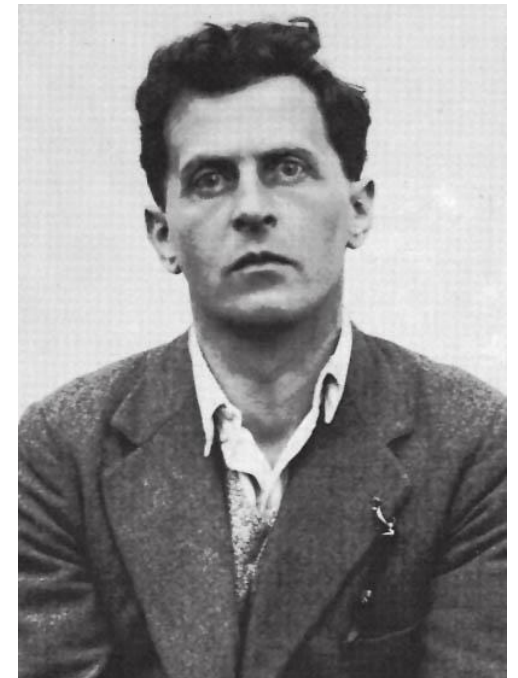
Skeptical of a formal theory of meaning definitions for each word

“The meaning of a word is its use in the language” - Philosophical Investigations.

Ludwig Josef Johann Wittgenstein
/'**ɪ**tgənʃtaɪn, -staɪn/
1889 –1951

Austrian-British Philosopher

Worked in logic, philosophy of mathematics, philosophy of mind, philosophy of language



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k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

How to ~~define~~ **capture** meaning of a word? **Distributional Hypothesis**

H_1 : Words occur in similar contexts tend to have similar meanings.

Like k -NN idea of birds of a feather flock together, but

It may or may not be correct!

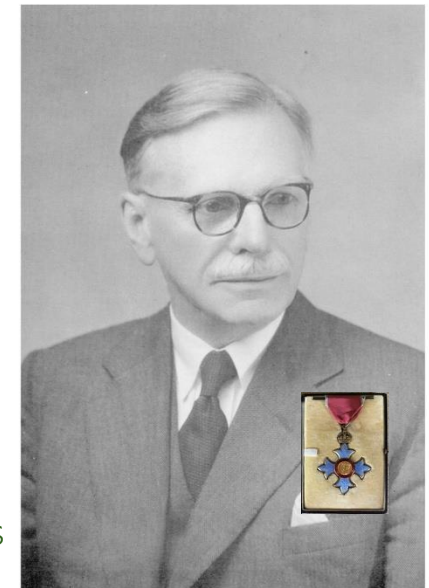
Joos, M. (1950). Description of language design.

Harris, Z. S. (1954). Distributional structure.

Firth, J. R. (1957). A synopsis of linguistic theory.

John Rupert Firth OBE (1890 – 1960)

Leading figure in British linguistics



JOHN RUPERT FIRTH

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k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

How to capture meaning of a word? H1: Distributional Hypothesis

“ongchoi delicious sauteed garlic ...”

“sauteed ongchoi ...”

context = 2 words

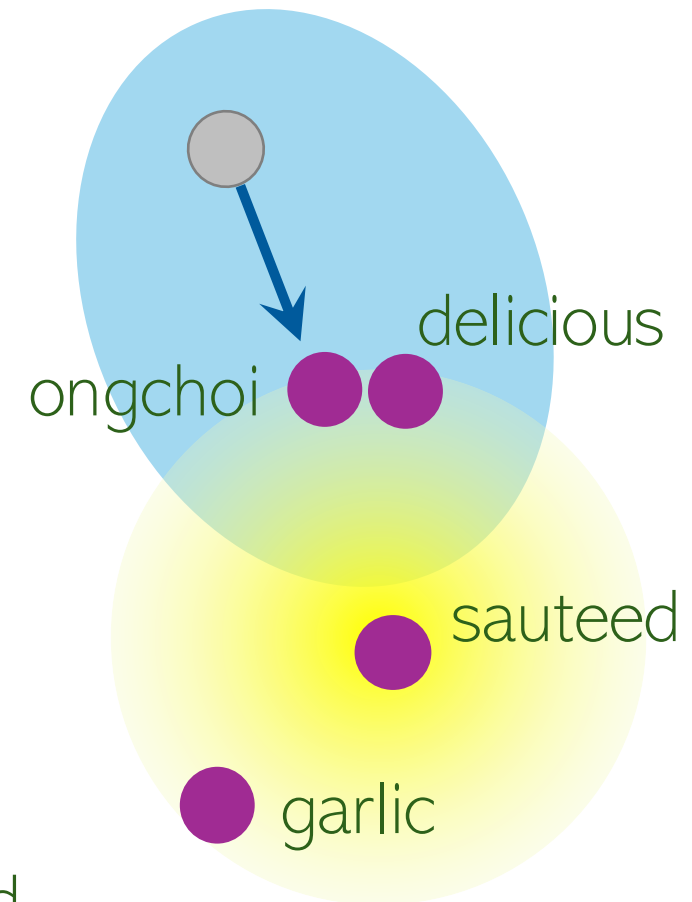
(ongchoi, delicious) = 1

(delicious, sauteed) = 1

(sauteed, garlic) = 1

(sauteed, ongchoi) = 1

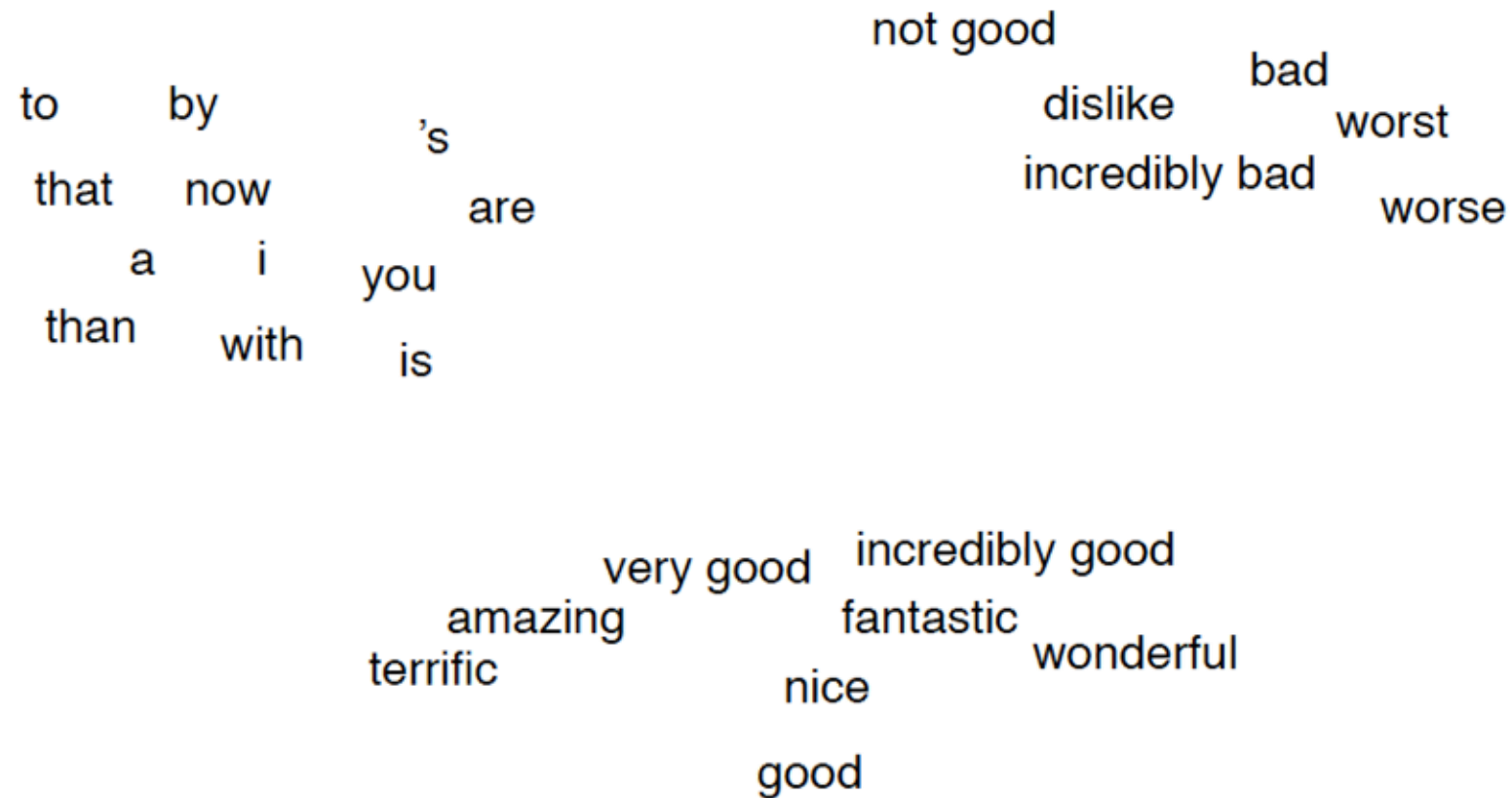
Ongchoi is not Person, nor Organization, but Food.



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k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

How to capture meaning of a word? H1: Distributional Hypothesis



Our mind is 2D Euclidian.

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k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=?}$

How to capture meaning of a word? H1: **Distributional Hypothesis**

Location of the words does *not* matter.

More cooccur in the same context

- More similar semantic
- Closer meaning
- Closer the distance!

Linguistic Semantic (Real-World) Proximity

- Cooccurrence Context in H1
- **Distributional Semantics**
- Geometric Proximity

Computational Leverage

not good
dislike
bad
worst
incredibly bad
worse

amazing
terrific
very good
incredibly good
fantastic
wonderful
nice
good

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k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

(1) Pick a text corpus (may or may not the same as $\mathbf{O}^*$)

(2) Define Context =

Fixed: Window of $\pm m$ words

Dynamic: Phrase, Sentence, Social Post, Paragraph, Page, Chapter, Book

(3) Count Word-Word Cooccurrences

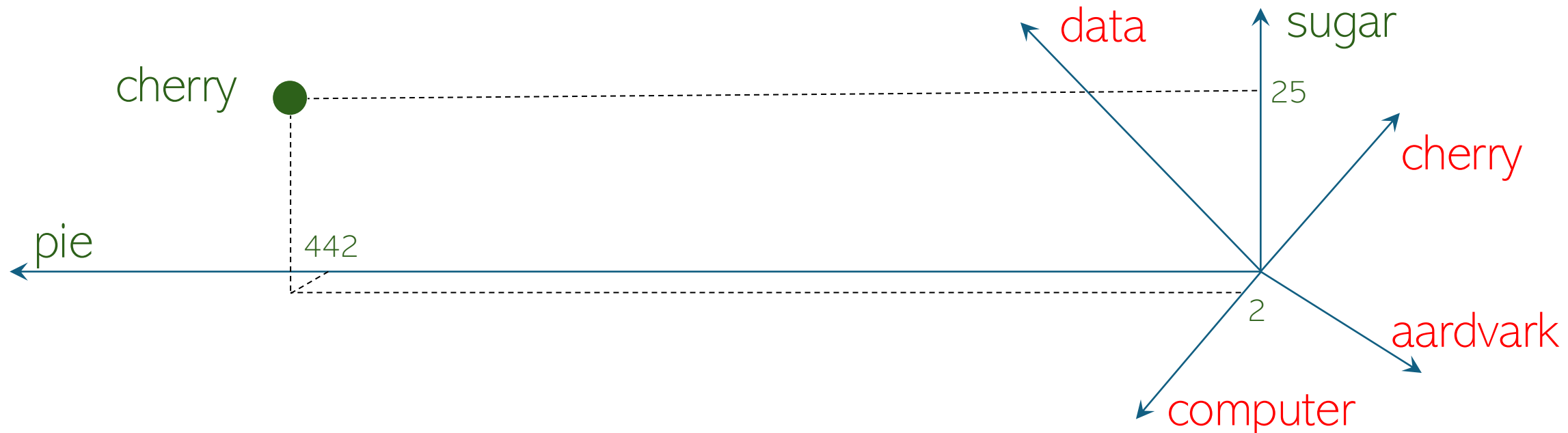
20

 k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|\mathbf{V}|}$

Corpus: Wikipedia articles
 Context: Window of ± 4 words

	aardvark	...	computer	data	result	pie	sugar	...
cherry	0	...	2	8	9	442	25	...

$1 \times |\mathbf{V}|$



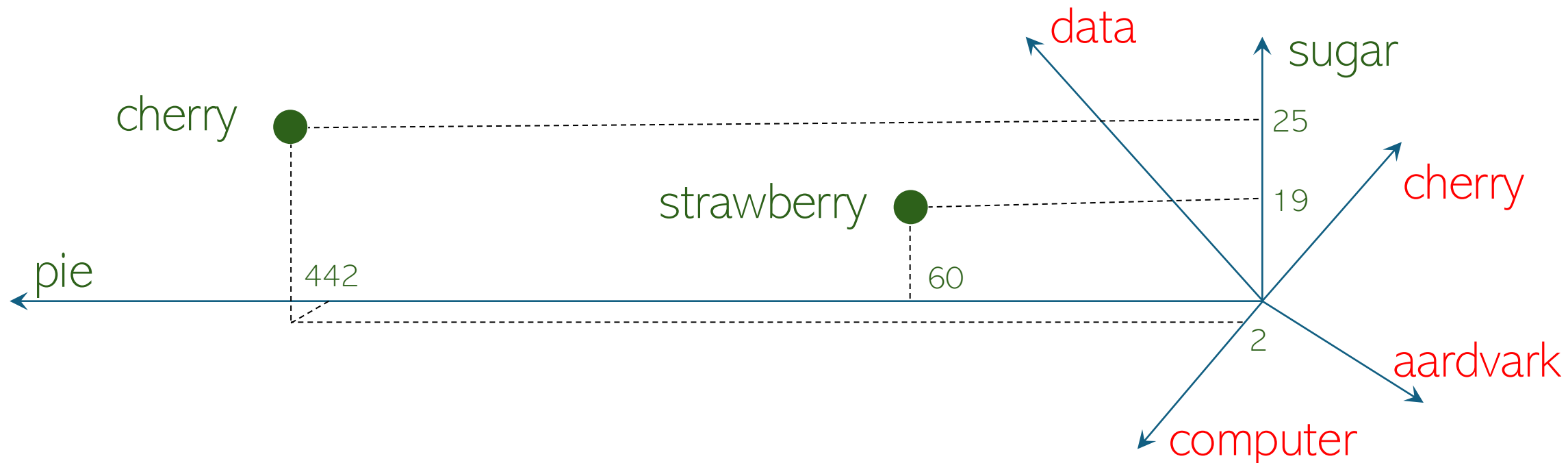
21

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

Corpus: Wikipedia articles
Context: Window of ± 4 words

	aardvark	...	computer	data	result	pie	sugar	...
cherry	0	...	2	8	9	442	25	
strawberry	0	...	0	0	1	60	19	

$1 \times |V|$
 $1 \times |V|$



22

 k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

Corpus: Wikipedia articles
 Context: Window of ± 4 words

	aardvark	...	computer	data	result	pie	sugar	...
cherry	0	...	2	8	9	442	25	
strawberry	0	...	0	0	1	60	19	
digital	0	...	1670	1683	85	5	4	
information	0	...	3325	3982	378	5	13	

 $1 \times |V|$ $1 \times |V|$ $1 \times |V|$ $1 \times |V|$

Word-Word Matrix
 Term-Term Matrix
 Word-Context Matrix

$$|V| \times |V|$$

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k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

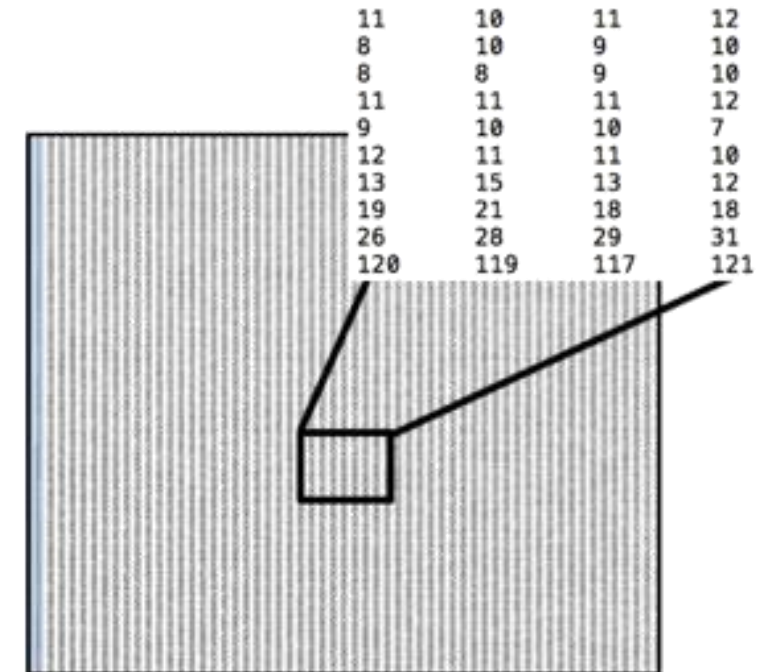
```
[6]
✓ Os 1 from collections import Counter, defaultdict
      2 docs = [
      3     """
      4     Ongchoi is a popular leafy green vegetable commonly used in
      5     Asian cuisine. Ongchoi has long hollow stems and tender green leaves.
      6     Many people cook ongchoi with garlic or add ongchoi to soups.
      7     """
      8 ]
      9
     10 context_size = 4
     11 word_word_matrix = defaultdict(Counter)
     12 for doc in docs:
     13     words = doc.split()
     14     for i, target in enumerate(words):
     15         # Context = words within ±context_size
     16         start = max(0, i - context_size)
     17         end = min(len(words), i + context_size + 1)
     18         for j in range(start, end):
     19             if i != j:
     20                 context_word = words[j]
     21                 word_word_matrix[target][context_word] += 1
     22
     23 vocab = sorted(set(word for doc in docs for word in doc.split()))# get vocabs
     24 print("      " + " ".join(f"{w:>7}" for w in vocab))
     25 for word in vocab:
     26     row = [word_word_matrix[word][w] for w in vocab]
     27     print(f"{word:<8}" + " ".join(f"{x:>7}" for x in row))
```


...	Asian	Many	Ongchoi	a	add	and	commonly	cook	cuisine.	garlic	green	has	hollow
Asian	0	0	1	0	0	0	1	0	1	0	0	1	0
Many	0	0	0	0	0	1	0	1	0	0	1	0	0
Ongchoi	1	0	0	1	0	0	0	0	1	0	0	1	1
a	0	0	1	0	0	0	0	0	0	0	1	0	0
add	0	0	0	0	0	0	0	0	0	1	0	0	0
and	0	1	0	0	0	0	0	0	0	0	1	1	1
commonly	1	0	0	0	0	0	0	0	1	0	1	0	0
cook	0	1	0	0	0	0	0	0	0	1	1	0	0
cuisine.	1	0	1	0	0	0	1	0	0	0	0	1	1

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k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

Word-Word Matrix: Sparse (majority of elements are zero) or Dense?



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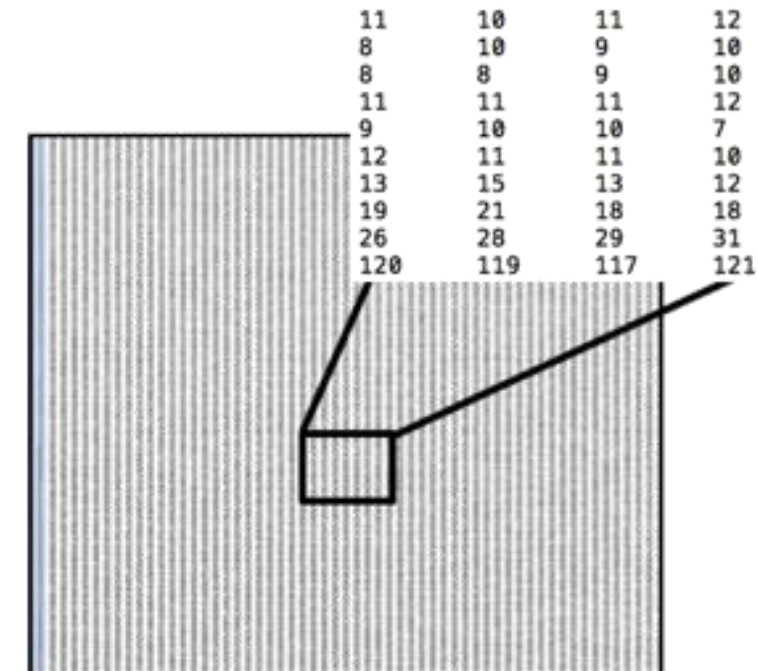
k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

Word-Word Matrix: Sparse or Dense? Sparse.

Since a word cooccur with few other words

Exceptions are words like: the, a, an, ...

$\rightarrow$ Stop words



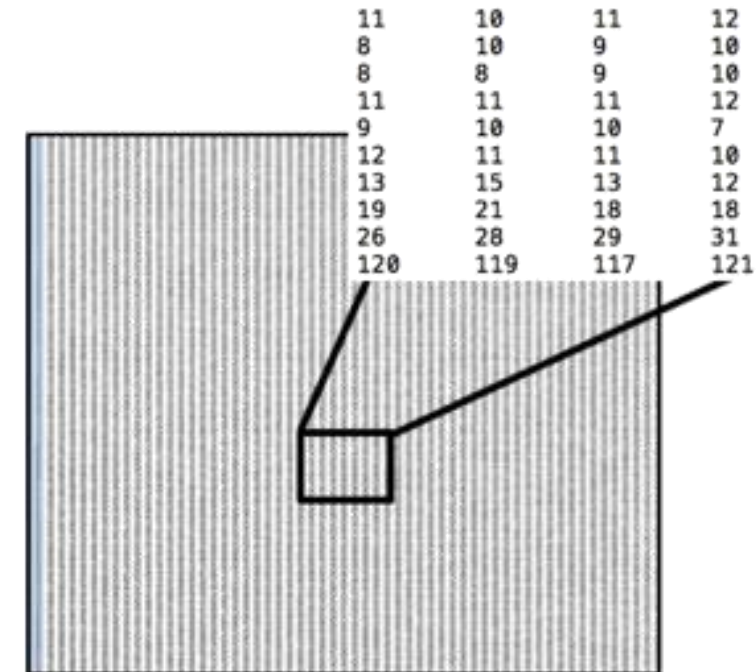
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k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

Small (± 1) or Large (± 1000) Context?

“water spinach or morning glory, ongchoi is a popular semi-aquatic”

“water spinach or morning glory, ongchoi is a popular semi-aquatic tropical”



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k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

Small (± 1) or Large (± 1000) Context? Should be tuned ...

The smaller the context,

The less chance two words cooccur

More sparsity, less noise in semantic similarity

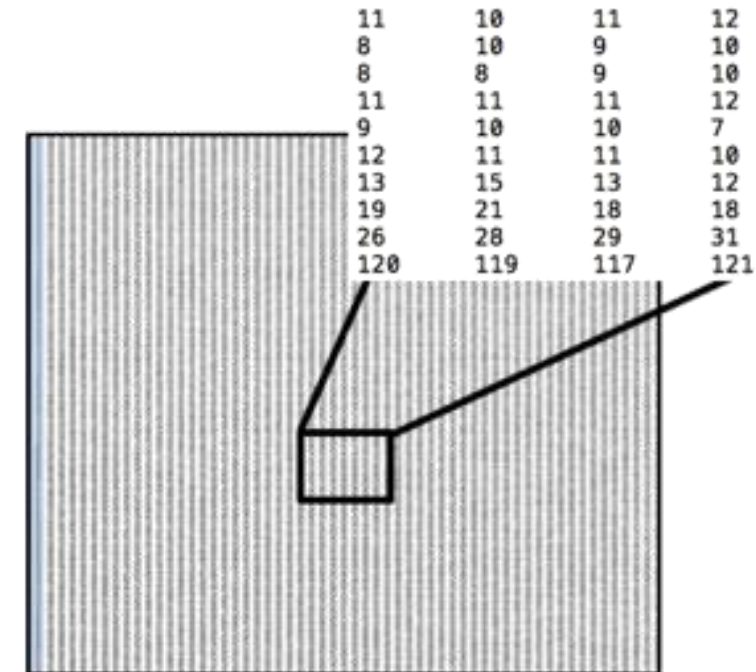
Conservative on semantic proximity

The larger the context,

The more chance two words cooccur

Less sparsity, more noise in semantic similarity

Liberal on semantic proximity



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k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

Word-Word (Term-Term) (Word-Context) Matrix

(1) Pick a text corpus (may or may not be the same as $\mathbf{O}^*$)

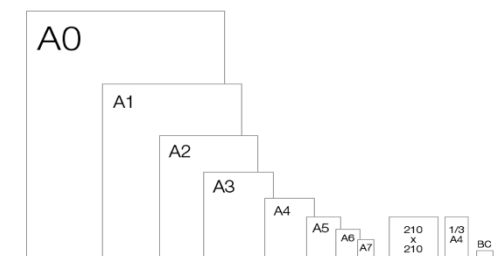
(2) Define Context =

Fixed: Window of $\pm m$ words

Dynamic: Phrase, Sentence, Social Post, Paragraph, Page, Chapter, Book

Document $\rightarrow$ a single line/file

(3) Count Word-Word Cooccurrences: all words in a line/file cooccur



k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

[13]

```

1  import pandas as pd
2  from sklearn.feature_extraction.text import CountVectorizer
3
4  docs = [
5      "ongchoi is a popular leafy green vegetable",
6      "ongchoi is commonly cooked with garlic",
7      "people eat ongchoi in soups and stir fries"
8  ]
9  count_model = CountVectorizer()# Document-Word matrix
10 X = count_model.fit_transform(docs)
11 Xc = X.T @ X# Document-level co-occurrence matrix
12 # Xc.setdiag(0)
13 words = count_model.get_feature_names_out()# Vocabulary
14 Xc_df = pd.DataFrame(Xc.toarray(), index=words, columns=words)
15 print(Xc_df)
16

```

v

	and	commonly	cooked	eat	fries	garlic	green	in	is	leafy
and	1	0	0	1	1	0	0	1	0	0
commonly	0	1	1	0	0	1	0	0	1	0
cooked	0	1	1	0	0	1	0	0	1	0
eat	1	0	0	1	1	0	0	1	0	0
fries	1	0	0	1	1	0	0	1	0	0
garlic	0	1	1	0	0	1	0	0	1	0
green	0	0	0	0	0	0	1	0	1	1
in	1	0	0	1	1	0	0	1	0	0
is	0	1	1	0	0	1	1	0	2	1

30

k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|\mathbf{V}|}$

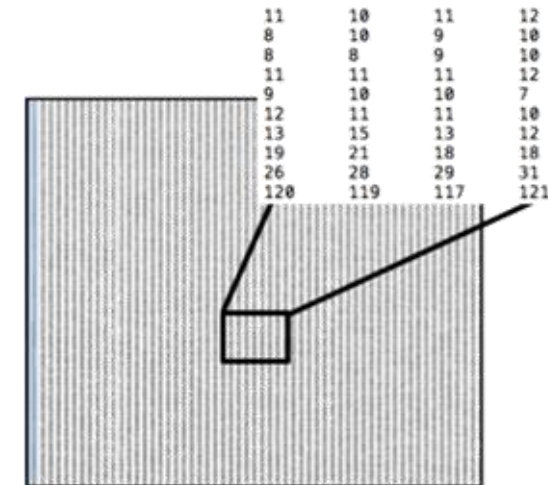
What happens to k -NN complexity? We already saw that $n \times |\mathbf{O}^*|$

$n = |\mathbf{V}| \simeq 520,000$ (size of online oxford dictionary)

k -NN $\rightarrow |\mathbf{V}| \times |\mathbf{O}^*|$

For 1 unseen observation: $|520,000| \times |\mathbf{O}^*|$

Can we reduce the $|\mathbf{V}|$ to a manageable size?



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k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

	aardvark	...	computer	data	result	pie	sugar	the	
cherry	0	...	2	8	9	442	25	205	$1 \times V $
strawberry	0	...	0	0	1	60	19	324	$1 \times V $
the	452	...	504	560	800	759	398	0	$1 \times V $

$$|V| \times |V|$$

Stop words, a static finite set of **non-informative** words, e.g., the, a, an, as
 $\rightarrow$ Filtering them shrinks the matrix horizontally/vertically



A friend to all
is a friend to none.

- Aristotle

Corollary by Hossein: "having such friend is not a value."

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 k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

	aardvark	...	computer	data	result	pie	sugar	find	
cherry	0	...	2	8	9	442	25	120	$1 \times V $
strawberry	0	...	0	0	1	60	19	524	$1 \times V $
find	452	...	504	560	800	759	398	0	$1 \times V $

 $|V| \times |V|$

But we need a dynamic list depending on the domain, e.g., lol, :), :D,

→ Dynamic set of **non-discriminative** words

→ Filtering them shrinks the matrix horizontally/vertically



A friend to all
is a friend to none.

- Aristotle

Corollary by Hossein: "having such friend is not a value."

33

 k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

	aardvark	...	computer	data	result	pie	sugar	... find
cherry	0	...	2	8	9	442	25	120 $\rightarrow$ 0
strawberry	0	...	0	0	1	60	19	524 $\rightarrow$ 0
find	452 $\rightarrow$ 0	...	504 $\rightarrow$ 0	560 $\rightarrow$ 0	800 $\rightarrow$ 0	759 $\rightarrow$ 0	398 $\rightarrow$ 0	0

$$|V| \times |V|$$

The more friend you have, the less valuable you are

The more cooccur with other words, the less accountable

(**find**, **aardvark**)=452 but also cooccur with many other words.

How many $\rightarrow$ Context Frequency (CF) $\rightarrow$ Row/Column Nonzero Counts $\rightarrow$ 6

Discount the value by CF $\rightarrow \frac{452}{CF=6} = 452 \times \frac{1}{6}$

inverse Context Frequency (i CF)

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 k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|V|}$

	aardvark	...	computer	data	result	pie	sugar	find	
cherry	$0 \times \frac{1}{4}$	...	$2 \times \frac{1}{6}$	$8 \times \frac{1}{6}$	$9 \times \frac{1}{6}$	$442 \times \frac{1}{6}$	$25 \times \frac{1}{6}$	$120 \times \frac{1}{6}$	$\rightarrow 6$
strawberry	$0 \times \frac{1}{4}$	...	$0 \times \frac{1}{4}$	$0 \times \frac{1}{4}$	$1 \times \frac{1}{4}$	$60 \times \frac{1}{4}$	$19 \times \frac{1}{4}$	$524 \times \frac{1}{4}$	$\rightarrow 4$
find	$452 \times \frac{1}{7}$	...	$504 \times \frac{1}{7}$	$560 \times \frac{1}{7}$	$800 \times \frac{1}{7}$	$759 \times \frac{1}{7}$	$398 \times \frac{1}{7}$	$0 \times \frac{1}{7}$	$\rightarrow 7$

$$CF(\mathbf{cherry})=6$$

$$CF(\mathbf{strawberry})=4$$

$$CF(\mathbf{find}, \mathbf{aardvark})=7$$

Discounts by
inverse Context Frequency

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k -Nearest Neighbor: Text: Word Space $\rightarrow \mathbb{R}^{n=|\mathbf{V}|}$

What happens to k -NN complexity? We already saw that $n \times |\mathbf{O}^*|$

$n = |\mathbf{V}| \simeq 520,000$ (size of online oxford dictionary)

k -NN $\rightarrow |\mathbf{V}| \times |\mathbf{O}^*|$

For 1 unseen observation: $|520,000| \times |\mathbf{O}^*|$

Can we reduce the $|\mathbf{V}|$ to a manageable size?

$n = |\mathbf{V}| \simeq 120,000$ (depending on the domain)

k -NN $\rightarrow |\mathbf{V}| \times |\mathbf{O}^*|$

For 1 unseen observation: $|120,000| \times |\mathbf{O}^*|$

Can we reduce the $|\mathbf{V}|$ to an **even more** manageable size?

